

MAST90084 Assignment X

Your Name (Student ID)

2026-04-04

```
# Reproducibility
library(conflicted)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
```

```
set.seed(1) # CHANGE TO YOUR STUDENT ID
```

Simple simulation + ggplot example

```
# Simulate basic data
n <- 100
x <- rnorm(n, mean = 0, sd = 1)
y <- 2 + 0.8 * x + rnorm(n, mean = 0, sd = 1)
sim_data <- tibble(x = x, y = y) |>
  arrange(x)

sim_data
```

```
# A tibble: 100 x 2
      x      y
  <dbl> <dbl>
1 -2.21 -0.422
2 -1.99  0.229
3 -1.80  0.301
4 -1.52 -0.686
5 -1.47  0.786
6 -1.38  1.20
7 -1.28  2.42
8 -1.25  0.663
9 -1.22  1.43
```

```
10 -1.13 0.167
# i 90 more rows
```

```
# Plot with ggplot2
ggplot(sim_data, ggplot2::aes(x = x, y = y)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Simulated data with linear trend",
    x = "x",
    y = "y"
  ) +
  theme_minimal()
```

``geom_smooth()`` using formula = 'y ~ x'



LaTeX examples (quick reference)

Use `equation` for one numbered equation.

$$\hat{\theta} = \arg \max_{\theta} \ell(\theta) \tag{1}$$

Use `split` inside `equation` when one equation is too long and needs line breaks, or has a flow of equals signs.

$$\begin{aligned}
\log L(\beta) &= \sum_{i=1}^n \left\{ y_i x_i^T \beta - \log(1 + e^{x_i^T \beta}) \right\} \\
&\quad - \frac{\lambda}{2} \beta^T \beta \\
&= \sum_{i=1}^n \left\{ y_i x_i^T \beta - \log(1 + e^{x_i^T \beta}) \right\} - \frac{\lambda}{2} \beta^T \beta
\end{aligned} \tag{2}$$

Use **gather** for multiple equations that are centered and not aligned at symbols.

$$\mathbb{E}[X] = \mu \tag{3}$$

$$\text{Var}(X) = \sigma^2 \tag{4}$$

Use **align** for multiple equations aligned at = (or another symbol via &).

$$\Pr(Y = 1 \mid x) = \frac{e^\eta}{1 + e^\eta}, \quad \text{for } \eta = x^T \beta \tag{5}$$

$$\log \left(\frac{\Pr(Y = 1 \mid x)}{1 - \Pr(Y = 1 \mid x)} \right) = x^T \beta \tag{6}$$

You can reference these using `Equation \ref{eq:single}`.

Question 1

(a)

Write your answer here.

```
# Code for Question 1(a)
```

(b)

Write your answer here.

```
# Code for Question 1(b)
```

Question 2

(a)

Write your answer here.

```
# Code for Question 2(a)
```

(b)

Write your answer here.

```
# Code for Question 2(b)
```

Question 3

(a)

Write your answer here.

```
# Code for Question 3(a)
```

(b)

Write your answer here.

```
# Code for Question 3(b)
```

Question 4

(a)

Write your answer here.

```
# Code for Question 4(a)
```

(b)

Write your answer here.

```
# Code for Question 4(b)
```